3D WEBSOFT INC. FOUNDATION

SCHOOL FUTURE TECH PROGRAM

Faculty Name :

Course Duration : 5 Months (150 Days)

Classes : Offline & Online

Daily Teaching : 3 Hours

Total Hours : 300 Hours

LEVEL 3 –Future Engineers(9th–10th Standard)

MODULE 1 – Python Foundations (Classes 1–10)

Class 4 – First Python Program

Complete Theory & Practical Notes

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Learning Objectives:-

By the end of this class, the session aims to:

1. Introduce students to the basic structure of a Python program.
2. Explain how Python instructions are written and executed.
3. Introduce the print() function and its purpose.
4. Teach students how to display text and numbers using Python.
5. Explain the importance of quotation marks when displaying text.
6. Introduce basic Python syntax and indentation awareness.
7. Teach students how to create, save, and run a Python program.
8. Help students understand the relationship between source code, execution, and output.
9. Develop students' confidence in writing their own Python statements.
10. Encourage students to experiment with simple code and observe the results.
11. Introduce basic programming errors and how to correct them.
12. Enable students to create a simple personalized “About Me” Python program.

1. What Students Will Learn

Core Concepts

- What is a Python program?
- What is source code?
- What is a statement?
- What is the print() function?
- Text/String values
- Numbers
- Quotation marks
- Python syntax
- Running a Python program
- Understanding program output
- Basic errors and debugging

Key Programming Flow

```
WRITE CODE
    ↓
SAVE PROGRAM
    ↓
RUN PROGRAM
    ↓
PYTHON EXECUTES CODE
    ↓
VIEW OUTPUT
```

Session

Topic	Type
Welcome & Recap	Discussion
What is a Python Program?	Theory
Understanding `print()`	Theory + Demonstration
Text, Numbers & Quotation Marks	Theory + Demonstration
Writing the First Python Program	Instructor Demonstration
Practical 1: Hello Python	Hands-on
Break / Coding Challenge	Activity
Practical 2: About Me Program	Hands-on
Multiple `print()` Statements	Practical
Errors & Debugging	Theory + Activity

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Topic	Type
Qui z + Exit Check	Assessment

2. Welcome & Recap

Begin with a quick recap of the previous classes:

- What is Python?
- What is VS Code?
- What is a .py file?
- What is the Python interpreter?
- How do we run Python code?

Quick Questions

"If Python is a language, what do we use to write that language?"

Expected answer:

VS Code / code editor

"What executes our Python code?"

Expected answer:

Python interpreter

Quick Demonstration

Open VS Code and show students:

```
Python_Foundations
        ↓
    Class04
        ↓
    first.py
```

Explain that today they will write their first real Python program**.**

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3. What Is a Python Program?

Theory

A **program** is a set of instructions given to a computer to perform a task.

A Python program contains instructions written using Python syntax.

For example:

```
print("Hello, Python!")
```

This tells Python:

Display the message **Hello, Python!**

Simple Analogy

Use the analogy of a **recipe**:

Recipe:

1. Take a cup.
2. Add water.
3. Add tea powder.
4. Boil.
5. Serve.

A computer program works similarly:

```
Instruction 1
      ↓
Instruction 2
      ↓
Instruction 3
      ↓
Computer performs the instructions
```

Important Vocabulary

Term	Meaning
Program	A set of instructions
Code	Instructions written by a programmer
Statement	A single instruction
Syntax	Rules for writing code

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Term	Meaning
Output	Result produced by a program
Interpreter	Executes Python code

4 .Understanding the print() Function

Theory

Introduce the first Python function:

```
print()
```

print() is used to display information as output.

Example 1

```
print("Hello")
```

Output:

```
Hello
```

Example 2

```
print("Welcome to Python")
```

Output:

```
Welcome to Python
```

Example 3

```
print(10)
```

Output:

```
10
```

Break down:

```
print("Hello")
```

into:

```
print → function
```

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```
( )   → parentheses
"Hello" → text/string
```

5. Text, Numbers & Quotation Marks

This is an important beginner concept.

A. Printing Text

```
print("Python")
```

Output:

```
Python
```

Explain that text is generally written inside quotation marks.

Examples:

```
print("Future Engineer")
print("I love coding")
print("Welcome to Python")
```

B. Printing Numbers

Numbers can be written without quotation marks:

```
print(10)
print(25)
print(100)
```

Output:

```
10
25
100
```

C. Compare

```
print("25") and print(25)
```

Explaination at beginner level:

- "25" → text
- 25 → number

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Mini Activity

Which ones are text?

```
print("Hello")
print(100)
print("100")
print("Python")
print(50)
```

Students identify the text and numbers.

6. Instructor Demonstration: First Python Program

"Now we are going to write our first Python program."

Step 1 – Open VS Code

Create/open the class folder:

```
Python_Foundations
```

Create:

```
Class04
```

Create a new file:

```
first_program.py
```

Step 2 – Write the Code

```
print("Hello, Python!")
```

Step 3 – Save

Save the file.

Step 4 – Run

Run the Python program using the VS Code Run button or terminal.

Expected Output

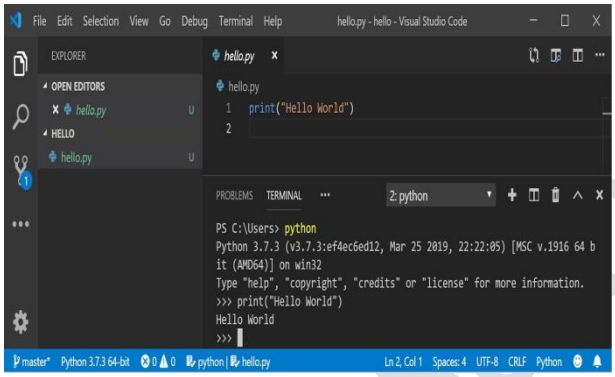
```
Hello, Python!
```

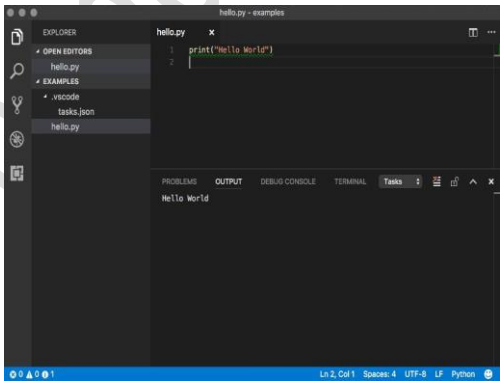
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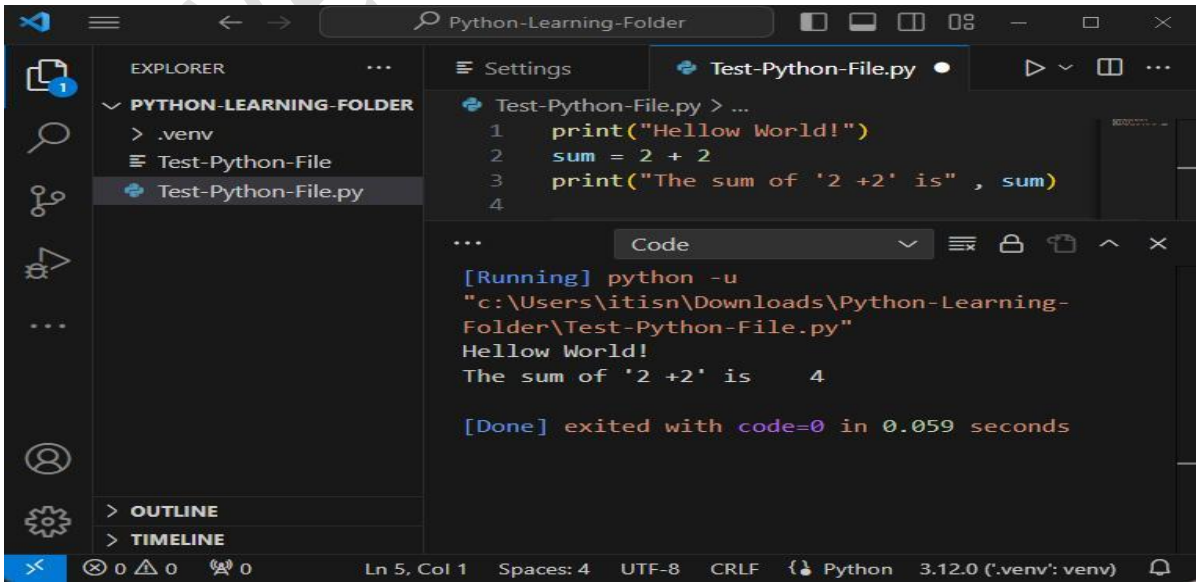
Explain the complete process:

```
first_program.py
      ↓
Python Code
      ↓
Python Interpreter
      ↓
Execution
      ↓
Output
```

6. Suitable Images / Visuals for This Section







Recommended classroom visuals

Use these visuals in the PPT:

1. **Python logo** – introduce the programming language.
2. **VS Code interface** – show where students write code.
3. **print() code screenshot** – connect theory to practice.
4. **Terminal output screenshot** – show input/code → output.
5. **Python execution flow diagram** – reinforce the programming process.

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8. Practical 1: Hello Python

Student Hands-on Activity

Students create:

```
first_program.py
```

and type:

```
print("Hello, Python!")
print("I am learning programming.")
print("Python is fun!")
```

Expected Output

```
Hello, Python!
I am learning programming.
Python is fun!
```

Students Tasks Checklist

Check whether students can:

- Open VS Code
- Create a .py file
- Type Python code
- Save the file
- Run the program
- Find the terminal/output
- Correct simple typing mistakes

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9. Coding Challenge

Give students a short challenge.

Challenge: Change the Message

Students modify:

```
print("Hello, Python!")
```

to something personal.

For example:

```
print("Hello, Future Engineers!")
```

Example:

```
print("Welcome to my program")
print("I am learning Python")
print("I will become a programmer")
```

Fun Challenge

"Can you make Python say something funny?"

Example:

```
print("My computer is my new friend!")
```

10.– Practical 2: My About Me Program

Students now create a small personalized program.

File Name

```
about_me.py
```

Starter Code

```
print("===== ABOUT ME =====")
print("My name is Alex")
print("I am a student")
print("I am learning Python")
print("I love technology")
print("My dream is to become an engineer")
```

Students replace the information with their own details.

Example Output

```
===== ABOUT ME =====
```

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```
My name is Rahul
I am a student
I am learning Python
I love technology
My dream is to become an engineer
```

Extension Challenge

Students can add:

```
print("My favorite subject is Science")
print("My favorite hobby is Drawing")
print("I want to build useful technology")
```

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11. Multiple print() Statements

Example:

```
print("My School")
print("My Name")
print("My Class")
print("My Hobby")
print("My Dream")
```

Python executes them **from top to bottom**.

Execution Flow

```
Statement 1
    ↓
Statement 2
    ↓
Statement 3
    ↓
Statement 4
    ↓
Statement 5
```

Demonstration

```
print("First")
print("Second")
print("Third")
```

Output:

```
First
Second
Third
```

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12. Errors & Debugging

Introduce the idea that programmers make mistakes.

Error Example 1

Correct:

```
print("Hello")
```

Incorrect:

```
print("Hello"
```

Error Example 2

Correct:

```
print("Python")
```

Incorrect:

```
print("Python")
```

Error Example 3

Correct:

```
print("Hello")
```

Incorrect:

```
Print("Hello")
```

Debugging Process

Teach students:

```
READ THE ERROR
      ↓
FIND THE LINE
      ↓
CHECK SPELLING
      ↓
CHECK QUOTATION MARKS
      ↓
CHECK PARENTHESES
      ↓
RUN AGAIN
```

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Mini Debugging Challenge

```
print("Hello Python)
print("I am learning Python"
print("Future Engineer")
```

13. Assessment & Exit Check

Quick Quiz

1. What is a Python program?

A. A set of instructions
B. A computer
C. A keyboard

Answer: A

2. Which function displays output?

A. `show()`
B. `print()`
C. `display()`

Answer: B

3. Which is correct?

A.

```
print("Hello")
```

B.

```
Print("Hello")
```

Answer: A

4. What is the output?

```
print("Python")
```

Answer:

```
Python
```

5. What is the purpose of quotation marks in:

```
print("Hello")
```

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14. Practical Assessment

my_first_program.py

```
print("==============================")
print("     MY FIRST PYTHON PROGRAM")
print("==============================")

print("My name is __________")
print("I am learning Python")
print("I love technology")
print("My favorite hobby is __________")
print("My dream is to become __________")
```

Success Criteria

A student successfully completes the task if they can:

Create a `.py` file
Write `print()` statements
Use quotation marks correctly
Save the program
Run the program
See the expected output
Fix at least one simple error independently

15. Homework / Extension Activity

Create a "My Future" Program

Students create:

`my_future.py`

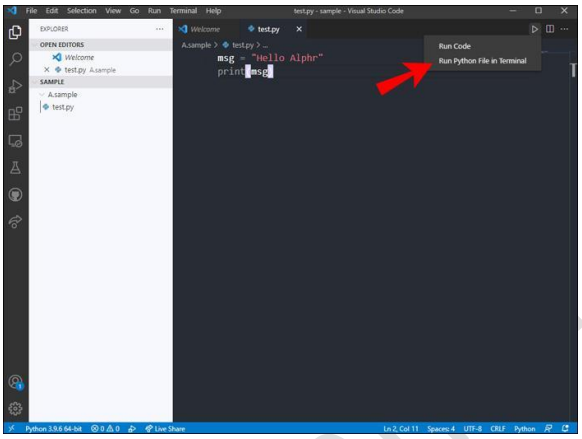
Example:

```
print("===== MY FUTURE =====")
print("My name is ______")
print("I am learning Python")
print("I want to become ______")
print("I want to build ______")
print("Technology can help me ______")
print("I will keep learning!")
```

Optional challenge: Add 5 more `print()` statements.

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16. Classroom Images





17. Teacher Materials

Required

- Computers/laptops
- Python installed
- VS Code installed
- Projector/display
- Internet connection if required
- Student notebooks

Software

Students should already have:

- Python
- VS Code

- Python extension for VS Code

For the current VS Code Python workflow, Microsoft recommends having the Python interpreter installed separately, the Python extension installed in VS Code, and the appropriate interpreter selected.

Learning Outcomes

By the end of this class, students will be able to:

1. **Explain** what a Python program is.
2. **Describe** the basic relationship between Python code, the interpreter, and program output.
3. **Identify** a Python statement and basic Python syntax.
4. **Explain** the purpose of the print() function.
5. **Write** simple Python print() statements.
6. **Display** text and numbers using Python.
7. **Use** quotation marks correctly when displaying text.
8. **Create and save** a Python .py file.
9. **Run** a Python program using VS Code.
10. **Observe and interpret** the output produced by a Python program.
11. **Write** multiple Python statements in the correct execution order.
12. **Identify and correct** simple syntax and spelling errors.
13. **Create** a personalized “About Me” or “My First Python Program.”
14. **Demonstrate confidence** in writing and executing a basic Python program independently.

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